

the two estimates has the form of a Oaxaca-Blinder-Kitagawa decomposition (Blinder 1973, Oaxaca 1973, Kitagawa 1955):

$$\hat{\beta}_{alt}^{DD} - \hat{\beta}^{DD} = \overbrace{s'(\hat{\beta}_{alt}^{2x2} - \hat{\beta}^{2x2})}^{\text{Due to 2x2 DDs}} + \overbrace{(s'_{alt} - s')\hat{\beta}^{2x2}}^{\text{Due to weights}} + \overbrace{(s'_{alt} - s')(\hat{\beta}_{alt}^{2x2} - \hat{\beta}^{2x2})}^{\text{Due to interaction}} \quad (18)$$

Dividing each term on the right side of (18) by $\hat{\beta}_{alt}^{DD} - \hat{\beta}^{DD}$ shows the proportional contribution of changes in the 2x2 DD's, changes in the weights, and the interaction of the two.³⁴ Differences arising from the 2x2 DD's show that control variables are correlated with treatment and the outcome, often pointing to a real sources of bias. Differences arising because of the weights, though, change the way the DD deals with heterogeneity and do not indicate that the 2x2 DD's are confounded. It is also simple to learn which terms drive each kind of difference by plotting $\hat{\beta}_{alt}^{2x2}$ against $\hat{\beta}^{2x2}$ and s against s_{alt} . One of the most valuable contributions of the DD decomposition theorem is to provide simple new tools for learning why estimates change across specifications.

i. Dropping Untreated Units

Papers commonly estimate models with and without untreated units, and the decomposition theorem shows that this is equivalent to setting all $s_{kU} = 0$ and rescaling the $s_{k\ell}$ to sum to one. Table 2 shows that this changes the unilateral divorce estimate so much that it becomes positive (2.42, s.e. = 1.81), but figure 6 suggests that this occurs not necessarily because of a problem with the design, but because *half* of the timing terms are biased by time-varying treatment effects.

ii. Population Weighting

Solon, Haider, and Wooldridge (2015) show that differences between population-weighted (WLS) and unweighted (OLS) estimates can arise in the presence of unmodeled heterogeneity, and suggest comparing the two estimators (Deaton 1997, Wooldridge 2001). WLS increases the

³⁴ Grosz, Miller, and Shenhav (2018) propose a similar decomposition for family fixed effects estimates.

influence of large *units* by weighting the means of y that make up each 2x2 DD, and it increases the influence of terms involving large *groups* by basing the decomposition weights on population rather than sample shares.

Weighting in the unilateral divorce analysis changes the DD estimate from -3.08 to -0.35. Table 2 indicates that just over half of the difference comes from changes in the 2x2 DD terms, 38 percent from changes in the weights, and 9 percent from the interaction of the two. Figure 9 scatters the weighted 2x2 DD's versus the unweighted ones. Most components do not change and lie along the 45-degree line, but large differences emerge for terms involving the 1970 states: Iowa and California.³⁵ Weighting, which obviously gives more influence to California, makes the terms that use 1970 states as treatments more negative, while it makes terms that use them as controls more positive. This is consistent either with an ongoing downward trend in suicides in California or, as discussed above, strongly time-varying treatment effects.³⁶

iii. Triple-Difference Estimator

When some units should not be (as) affected by a given treatment, they can be used as a falsification test. Assume that units belong to either an affected group ($G_i = 1$) or an unaffected group ($G_i = 0$). The simplest way to incorporate the “third difference”, G_i , would be to estimate separate DD coefficients in each sub-sample: $\hat{\beta}_0^{DD}$ and $\hat{\beta}_1^{DD}$. One could equivalently estimate the

³⁵ Lee and Solon (2011) observe that California drives the divergence between OLS and WLS estimate in analyses of no-fault divorce on divorce rates (Wolfers 2006).

³⁶ Weighting by (a function of) the estimated propensity score (Abadie 2005) is often used to impose covariate balance between treated and untreated units (see Bailey and Goodman-Bacon 2015). The decomposition theorem points to two limitations of this approach. First, reweighting untreated observations has no effect on the timing terms. Second, reweighting untreated observations using the estimated probability of being in *any* treatment group does not impose covariate balance within each pair. By changing the relative weight on different untreated units but leaving their total weight the same, this strategy does not change s , so all differences stem from the way reweighting affects the $\hat{\beta}_{kU}^{2x2}$ terms. Table 2 estimates reweighted models based on a propensity score equation that contains the 1960 sex ratio and per-capita income from figure 8, as well as the 1960 general fertility rate and infant mortality rate. This puts much more weight on Delaware and less weight on New York, and makes almost all $\hat{\beta}_{kU}^{2x2}$ much less negative, changing the overall DD estimate to 1.04. Callaway and Sant'Anna (2018) propose a generalized propensity score reweighted estimator for DD models with timing variation.

following triple-difference model (DDD) on the pooled sample including interactions of G_i with all variables from equation (2):³⁷

$$y_{it} = \alpha_i + \alpha_t + \beta_0^{DD} D_{it} + \alpha_t G_i + \beta^{DDD} D_{it} G_i + e_{it} \quad (19)$$

$\hat{\beta}_0^{DD}$ is the two-way fixed effects DD estimate for the $G_i = 0$ sample and $\hat{\beta}^{DDD}$ equals the difference between the sub-sample DD coefficients: $\hat{\beta}_1^{DD} - \hat{\beta}_0^{DD}$.

One problem with this estimator is that $\hat{\beta}_1^{DD}$ equals an average weighted by the cross-sectional distribution in the $G_i = 1$ sample, but $\hat{\beta}_0^{DD}$ uses the cross-sectional distribution of the $G_i = 0$ sample. If G_i were an indicator for black respondents, then 2x2 DD's that included Southern states would get more weight in the black than the white sample while the opposite would be true for Vermont and New Hampshire. Estimates of (19) difference out cross-state/cross-year changes in white outcomes weighted by white populations, and so may not capture relative trends by race *within* states. A null result for $\hat{\beta}_0^{DD}$, which is typically reassuring, could be driven by completely different 2x2 DD's than the ones that matter most for $\hat{\beta}_1^{DD}$.

DDD specifications that include a more saturated set of fixed effects overcome this problem. If treatment rolled out by state (s), for example, a DDD model can include state-by-time fixed effects ($\alpha_s \alpha_t$):

$$y_{it} = \alpha_i + G_i \alpha_s + G_i \alpha_t + \alpha_s \alpha_t + \beta^{DDD} D_{it} G_i + e_{it} \quad (20)$$

The DDD estimate from (20) does equal a weighted average of 2x2 DD's. $\hat{\beta}^{DDD}$ is equivalent to first collapsing the data to mean differences between G -groups within (s, t) cells, $\bar{y}_{st}^{G=1} - \bar{y}_{st}^{G=0}$, then estimating a DD model weighted by cell sizes times the $\widehat{var}(G_i|s, t) = g_{st}(1 - g_{st})$, where

³⁷ In this set up the third difference partitions units, so $\alpha_i G_i$ is collinear with α_i .

g_{st} is the mean of G_i by s and t . Unlike (19), estimates from (20) *do* net out changes across G_i within state/year cells. This changes the weights, though, because the introduction of variation across the third difference within a cell leads to the typical OLS result that cells with more variation get more weight. In this case, “more variation” in sample membership within a cell means approximately equal numbers of units with $G_i = 1$ and $G_i = 0$.³⁸

Recasting this version of a DDD model as a DD on differences by G_i implies that the DD decomposition theorem holds, albeit with a slight change to the calculation of the weights. All the results and diagnostic tools derived above apply to specifications like (20) by defining the outcome as $\bar{y}_{st}^{G=1} - \bar{y}_{st}^{G=0}$ and using the proper weights.

iv. Unit-specific linear time trends

Researchers control for unit-specific linear time trends to allow “treatment and control states to follow different trends” (Angrist and Pischke 2009, p 238), and view it as “an important check on the causal interpretation of any set of regression DD estimates” (Angrist and Pischke 2015, p. 199). Appendix B derives a closed-form solution for the detrended estimator and shows that a version of the DD decomposition theorem applies to it. The specific way that unit-specific trends change each 2x2 component fit with previous intuition. They essentially subtract the cross-group difference in averages of y before and after the middle of the panel, $\bar{t}$, but these differences are weighted by absolute distance to $\bar{t}$: $|t - \bar{t}|$. This is, as Lee and Solon (2011) point out, akin to a regression discontinuity design in that the estimator relies less on variation at the beginning or end of the panel because this variation is absorbed by the trends. Unfortunately, trends tend to absorb time-varying treatment effects that are necessarily larger at the end of the panel, and in these cases

³⁸ Collapsing the data to the within-cell mean differences $\bar{y}_{st}^{G=1} - \bar{y}_{st}^{G=0}$ and running OLS on the aggregated data (or WLS using cell populations) would eliminate the $g_{st}(1 - g_{st})$ from the decomposition weights.